

# When Does Action-Type Supervision Help GUI Grounding? Separating Real Effects from Evaluation Artifacts

Aadi Chauhan  
Stanford University  
aadic@stanford.edu

Arthur Ilyasov  
Stanford University  
ailyasov@stanford.edu

## Abstract

GUI agents must predict an action type (click, scroll, type, ...) and a spatial target from a screenshot and an instruction. We study whether explicitly factoring these decisions helps a 2B-parameter agent, via a matched-compute comparison of five conditioning mechanisms on Qwen2-VL: a flat baseline (A), an auxiliary action-type loss (B), hard routing (C), an additive embedding hook (D-hook), and the prepended learned action embedding our original proposal advocated (D-token). Equally central, we report how our own first evaluation of this question misled us in three distinct ways, each of which we diagnose, quantify, and repair: (i) injecting conditioning via inputs\_embeds silently bypasses Qwen2-VL’s M-RoPE position path and manufactured an  $8\sigma$  false negative; (ii) train/validation splits taken as consecutive slices of one episode stream leaked shared task goals (24% of validation steps at the headline setting, up to 85% at scale); and (iii) a paired bootstrap that pooled (seed, example) predictions as independent units was anti-conservative, rejecting a true clustered null 21.2% of the time at nominal 5% (an episode-clustered replacement achieves 5.1%). After the statistical repair, the auxiliary-loss effect survives on the historical data (B–A hit@0.10 +0.051 [+0.020, +0.081], permutation  $p = 0.0043$ , versus the invalid pooled  $p = 0.0002$ ); we then retrain all variants on frozen goal-disjoint splits and evaluate once on an untouched test set under a prespecified analysis. On that test set, B–A is [PENDING] (Holm  $p =$  [PENDING]) and the end-to-end predicted-type pipeline versus A is [PENDING] (Holm  $p =$  [PENDING]). The mechanism our proposal bet on is refuted as the winner: D-token is causally used at inference (wrong or zeroed embeddings severely hurt) yet never beats the plain auxiliary loss, and exploratory per-class decompositions locate the gain in click-vs-scroll/type disambiguation rather than field localization. We release the frozen splits, per-example predictions, and the clustered-inference code; we argue that offline step-level grounding studies at this scale are unusually easy to contaminate, and that the three

artifacts we document are likely present in other evaluations.

## 1. Introduction

A GUI agent maps a screenshot and a natural-language instruction to a low-level action: a type (click, scroll, type, drag, ...) and, for spatial actions, a pixel-level target. State-of-the-art native agents (UITARS [13], OS-Atlas [11], SeeClick [4], ShowUI [7]) emit both in one autoregressive stream, entangling the two decisions. A documented failure mode follows: commit to the wrong action type, then ground coherently within the wrong semantic frame – scrolling past a visible target instead of clicking it. This motivated our original hypothesis, formulated as an architecture bet: a learned action-type embedding, prepended to the instruction, should let a grounding model condition on what before deciding where, and should help most in low data.

This paper reports what happened when we tested that bet carefully, and it has two intertwined subjects. The first is the scientific question: when does action-type supervision change GUI grounding, and through which mechanism? The second is methodological: which implementation and evaluation choices can create a false appearance of benefit or harm in studies of exactly this kind? We came to the second subject involuntarily – three separate artifacts in our own pipeline each produced a wrong conclusion at some point during the project – and we believe the post-mortem is at least as useful to practitioners as the object-level result.

Concretely, we compare five matched-compute Stage-2 conditioning mechanisms on Qwen2-VL-2B [10] with LoRA, over AITW [9] with a Mind2Web [5] cross-dataset control, preceded by a Stage-1 action-type classifier study. Our contributions:

- A controlled answer to the factorization question. Auxiliary action-type supervision (B) improves grounding where action type is spatially predictive; the effect vanishes on a matched con-

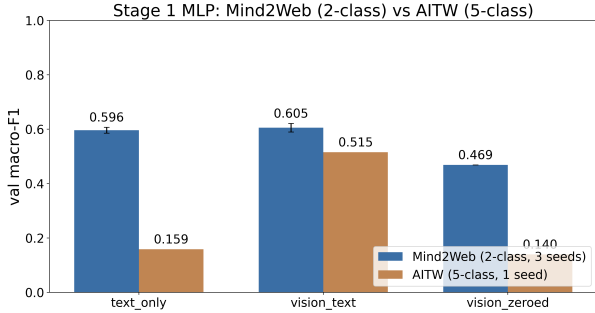


Figure 1. Stage 1 action-type classifier macro-F1 on Mind2Web (2 active classes) and AITW (5 active classes). Vision is essential on AITW (+0.414 over text-only) but redundant on Mind2Web (+0.009), because AITW instructions describe goals rather than physically specifying the action.

trol where it is not. The proposal’s specific mechanism – a routable prepended embedding (D-token) – is causally used by the trained model yet never outperforms the far simpler auxiliary loss, refuting the architecture bet while supporting the broader thesis.

- Three quantified evaluation pitfalls. (1) A silent M-RoPE / inputs\_embeds interaction that fabricated an  $8\sigma$  false negative; (2) episode-stream train/validation leakage that is invisible at the episode level (2% of validation steps) but large at the shared-task-goal level (24%–85%); (3) pseudo-replicated significance testing whose anti-conservativeness we measure on synthetic clustered nulls (21.2% type-I error at nominal 5%).
- A repaired, releasable protocol. Pinned-revision, goal-disjoint frozen train/validation/test manifests with an untouched test region beyond every historical data access; episode-clustered paired inference with per-seed effects and validated calibration; and a prespecified confirmatory analysis committed before test access. All artifacts, per-example predictions, and analysis code are released.
- An honest confirmatory result. We report the prespecified untouched-test outcome for every primary contrast whatever its direction, alongside the exploratory history that motivated it.

## 2. Related Work

End-to-end GUI agents. The dominant paradigm treats action type, arguments, and coordinates as one

autoregressive output. UI-TARS [13] exemplifies this with a unified action space across platforms and explicit thought traces. OS-Atlas [11] scales the same paradigm with 13M cross-platform GUI grounding instances and shows that action-label harmonization across web/Android/desktop is nontrivial. ShowUI [7] builds a compact VLA around Qwen2-VL with interleaved vision-language-action streaming. Our flat baseline (Variant A) replicates the joint-decoding paradigm of these systems.

Modular grounding. SeeClick [4] argues that grounding is the central bottleneck in visual GUI agents. UGround / SeeAct-V [6] separates planning from grounding at the text-vs-pixel boundary. GUI-Actor [14] replaces coordinate-token regression with an attention-based action head driven by a dedicated <ACTOR> token plus a verifier. Our work studies a different module boundary: between action-type prediction and spatial grounding, not between planning text and coordinates.

Reward decomposition. UI-R1 [8] applies RL fine-tuning with a rule-based reward that separates action-type accuracy from action-argument accuracy on AndroidControl. It is the closest training analog: type and location as separable optimization targets; we use supervised conditioning rather than RL. Region-Focus [2] refines grounding at inference; we change the training-time conditioning pathway. SayCan [1] and ReAct [12] provide the modular-decision precedent, and RT-2 [3] the unified token-level action generation paradigm we critically examine.

Evaluation integrity. Our methodological findings echo, in a GUI setting, long-standing concerns about test-set contamination and dependence-blind inference: grouped data require grouped splits and cluster-aware uncertainty, and repeated measurements of the same example must not be pooled as independent evidence. Mind2Web’s own metric design (element accuracy vs. operation F1 vs. step success) [5] reflects the type/location decomposition we study.

## 3. Data and Splits

### 3.1. Sources and preprocessing

We use two datasets under a canonical 8-class action taxonomy (click, double\_click, type, scroll, drag, hotkey, wait, finished).

AITW [9] (via the cjfcsjt/AITW\_General Hugging-Face mirror, revision pinned to 5c0dc713) is the primary Stage-2 benchmark: five active classes (click 55%, scroll 13%, finished 12%, type 11%, hotkey 9%). Screenshots in the mirror are raw RGB bytes without headers; we decode them against seven observed resolu-

tions with a portrait-aspect fallback. DUAL\_POINT events are split into taps (click) versus swipes (scroll) by normalized touch–lift distance (threshold 0.04). Two data mixes matter below: all\_with\_coords (tap, swipe, type – action type is spatially predictive because taps target small elements while swipes are gestures) and taps\_and\_swipes (a control where knowing the type carries little information about where).

Mind2Web [5] (multimodal release) provides the Stage-1 cross-dataset comparison and a Stage-2 cross-dataset control; it is click/type-dominated with only two active classes at scale.

Coordinates. Targets are normalized to  $[0, 1]^2$  and rendered as the string "(x, y)" on a  $[0, 999]$  grid; training loss is masked to the coordinate tokens.

### 3.2. Two evaluation protocols

Exploratory protocol (Phases 1–7, historical). Our original runs streamed the AITW training split in mirror order, filtered by data mix, and took consecutive slices: the first  $n_{\text{train}}$  steps for training and the next  $n_{\text{val}}$  for validation. This is the kind of split that looks innocuous in code review and is not: episodes and, worse, repeated task goals straddle the boundary (Section 5.2). Every historical number in this paper is labeled exploratory and is re-analyzed with dependence-aware statistics.

Confirmatory protocol (Phase 9, frozen). We rebuilt the data pipeline around a frozen manifest: the first 12,000 all\_with\_coords steps of the pinned mirror revision, grouped by normalized task goal (episodes never span goals), split goal-disjointly into train (9,511 steps), validation (1,660), and an untouched test region (829): test groups are drawn exclusively from beyond stream index 5,500, past the furthest index any historical run ever loaded (5,250), so no test step was trained on, evaluated on, or inspected during the exploratory phase. Evaluation sets are frozen and identical for every variant and training size (250 validation / 600 test steps for all\_with\_coords; 200/400 for the control mix); training subsets are nested prefixes. The manifest, split hashes, 17 disclosed near-duplicate goal pairs (token Jaccard  $\geq 0.8$ ), duplicate checks, and the prespecified analysis (contrasts, metrics, corrections, failed-run rules) were committed to version control before any test-set evaluation ran. We claim "decisions frozen before test access," documented by commit history – not third-party preregistration.

## 4. Methods

### 4.1. Stage 1: action-type classifier

A frozen Qwen2-VL-2B encodes screenshot+instruction; mean-pooled final hidden states (1536-d) feed a 3-layer MLP trained with class-weighted cross-entropy (AdamW, lr  $10^{-4}$ , batch 128, 8 epochs, best epoch by validation macro-F1). Modes: text\_only vs. vision\_text.

### 4.2. Stage 2: five conditioning mechanisms

Stage 2 fine-tunes Qwen2-VL-2B with LoRA (rank 16 on q/k/v/o projections,  $\approx 0.2\%$  of parameters) to emit the coordinate string; all variants share data, optimizer (lr  $2 \times 10^{-5}$ , 2 epochs, batch 1), and evaluation. The only difference is how action-type information enters:

A (flat baseline). No conditioning; replicates joint decoding.

B (auxiliary loss). A linear action-type head on the pooled representation adds cross-entropy to the grounding loss ( $\lambda=1$ ); no action information at inference.

C (hard routing). Trained to emit "{word} (x, y)"; at inference the gold action word is forced as the decode prefix.

D-hook (additive embedding). A learned table  $E \in \mathbb{R}^{8 \times d}$ ; the row for the current action type is added to all token embeddings via a forward hook, preserving the normal input\_ids path (embedding norm at convergence  $\approx 0.07$  at the headline setting).

D-token (prepended token – the original hypothesis). A sentinel  $\langle \text{action\_slot} \rangle$  token is prepended to the instruction and its embedding replaced in place by  $E[a]$ , M-RoPE-correctly (the token is in input\_ids); norm at convergence  $\approx 2.2$ , a strong routable signal.

Objective. With screenshot  $s$ , instruction  $g$ , action type  $a$ , coordinate tokens  $y_{1:T}$  and variant-specific conditioning  $c$ :

$$\mathcal{L}_{\text{coord}} = - \sum_{t=1}^T \log p_{\theta}(y_t \mid y_{<t}, s, g, c), \quad (1)$$

with  $c = \emptyset$  for A and B,  $c = E[a]$  for D-hook/D-token, and the forced word for C; B adds  $\lambda \text{CE}(h_{\phi}(\bar{z}), a)$  (Eq. 1 otherwise unchanged).

End-to-end pipeline. The deployable configuration trains D-hook, then a small MLP head on frozen-backbone features (Stage 1 in-job), and evaluates grounding under predicted types; gold-type results are oracle ceilings, never deployable claims.

### 4.3. Evaluation metrics

hit@ $r$ : fraction of predictions within normalized L2 distance  $r$  of the target ( $r \in \{0.05, 0.10, 0.25\}$ ; headline 0.10). Mean L2: mean normalized distance (parse failures scored at the  $\sqrt{2}$  sentinel and counted as misses; parse rates reported). Statistical methodology is Section 5.3.

## 5. Three Evaluation Pitfalls, Quantified

### 5.1. Pitfall 1: M-RoPE vs. inputs\_embeds

Qwen2-VL assigns 3D (temporal, height, width) position IDs to visual tokens via an input\_ids-keyed path. Our first D-token implementation injected the action embedding by passing inputs\_embeds directly, which silently bypasses that path: image-token positions were computed without the action slot in the sequence, misaligning multimodal position embeddings for every visual token. The observed effect was a confident  $8\sigma$  “A beats D” result – a pure artifact. Rebuilding the same mechanism through the input\_ids path (D-hook, D-token) erased the deficit. Any conditioning study on M-RoPE-family models that touches inputs\_embeds should verify position-ID computation explicitly; the artifact is silent, large, and directionally misleading.

### 5.2. Pitfall 2: stream-slice splits leak task goals

Table 1 quantifies the historical protocol’s leakage by reconstructing, for every historical cell, which episodes and normalized task goals each validation step shared with that cell’s training slice. Episode-level overlap – the leakage one usually checks – is small (2% at the headline cell). The dominant channel is shared task goals: 24% of headline validation steps have a goal string that also appears in training, rising to 85% at  $n_{\text{train}}=5000$  – AITW goals repeat across many episodes, so consecutive-slice splitting almost guarantees template overlap at scale. Val-set difficulty therefore varies with  $n_{\text{train}}$  (each size had a different val slice), quietly confounding the historical scaling curve. The confirmatory protocol removes both channels by construction (goal-disjoint splits; Section 3.2).

### 5.3. Pitfall 3: pseudo-replicated significance

Our original headline significance pooled per-example distances across seeds – treating  $3 \times 250$  (seed, example) rows as 750 independent paired observations. Predictions of the same example by different seeds are strongly correlated, and examples within an episode share state; the pooled bootstrap therefore overstates evidence. On synthetic nulls matching the data’s dependence structure (episode and example effects shared

| historical cell       | val steps | sharing a train episode | sharing a tr |
|-----------------------|-----------|-------------------------|--------------|
| all_wc, $n=300$       | 250       | 4 (2%)                  | 14 (6%)      |
| all_wc, $n=500$       | 250       | 4 (2%)                  | 47 (19%)     |
| all_wc, $n=800$       | 250       | 5 (2%)                  | 49 (20%)     |
| all_wc, $n=1200$      | 250       | 4 (2%)                  | 61 (24%)     |
| all_wc, $n=2500$      | 250       | 10 (4%)                 | 133 (53%)    |
| all_wc, $n=5000$      | 250       | 2 (1%)                  | 213 (85%)    |
| taps_swipes, $n=1000$ | 200       | 0 (0%)                  | 48 (24%)     |

Table 1. Leakage in the historical (exploratory) splits: fraction of validation steps sharing an episode or a normalized task goal with that cell’s training slice. Goal sharing, not episode straddling, is the dominant channel.

across seeds; zero true difference), the pooled test rejects 21.2% of the time at nominal 5% (1000 simulations), while the replacement below achieves 5.1%.

Repaired inference. For each contrast we (i) average each example’s paired difference over seeds, (ii) cluster examples by episode, (iii) bootstrap clusters (10,000 resamples) for 95% CIs, and (iv) use an episode-level sign-flip permutation test (10,000 permutations;  $p$ -values reported no smaller than Monte Carlo resolution – never  $p=0$ ). Estimand. This tests the fixed trained checkpoints on new data from the eval population. It is not calibrated for claims about the training procedure: adding run-level (seed) offsets to the synthetic null drives the episode-clustered test to 55% rejection (and the pooled test to 70%). With only three seeds per variant we therefore report per-seed means everywhere and make no procedure-level significance claims. The retracted pooled  $p$ -values (including a headline  $p=0.0002$ ) should not be cited.

## 6. Exploratory Results (historical splits, repaired statistics)

All numbers in this section come from the leaky historical splits and are therefore exploratory; they are shown with episode-clustered statistics, and Section 7 gives the prespecified untouched-test versions of every primary claim.

### 6.1. Stage 1: vision is essential exactly where instructions under-specify the action

Table 2 and Figure 1: on Mind2Web, instructions name the action (“click the submit button”), so vision adds +0.009 macro-F1 over text; on AITW, instructions state goals (“open a new Chrome private window”), text-only collapses toward the majority class, and vision adds +0.414. A TF-IDF ceiling of 0.168 on AITW confirms text is uninformative there regardless of classifier strength. This asymmetry decides where

| Variant         | Mind2Web<br>macro-F1                | AITW<br>macro-F1                    |
|-----------------|-------------------------------------|-------------------------------------|
| text_only       | 0.597 $\pm$ 0.011                   | 0.141 $\pm$ 0.032                   |
| vision_text     | <b>0.605 <math>\pm</math> 0.016</b> | <b>0.555 <math>\pm</math> 0.036</b> |
| vision_zeroed   | 0.469 $\pm$ 0.000                   | 0.110 $\pm$ 0.051                   |
| Vision delta    | +0.009                              | +0.414                              |
| TF-IDF + LogReg | 0.622                               | 0.168                               |
| Majority floor  | 0.472                               | 0.140                               |

Table 2. Stage 1 action-type classification (3 seeds, mean $\pm$ std; exploratory splits). On Mind2Web the instruction text alone contains the action type; on AITW it does not, and vision contributes +0.414 macro-F1. vision\_zeroed collapsing to the majority floor is the sanity check that features, not priors, drive the result.

| variant             | hit@0.10          | mean L2 $\downarrow$ | $\Delta$ hit@0.10 vs. A [95% CI] | perm. $p$ |
|---------------------|-------------------|----------------------|----------------------------------|-----------|
| A (flat)            | 0.255 $\pm$ 0.021 | 0.392                |                                  |           |
| B (aux loss)        | 0.305 $\pm$ 0.012 | 0.362                | +0.051                           |           |
| C (hard routing)    | 0.279 $\pm$ 0.032 | 0.412                | +0.024                           |           |
| D-hook (additive)   | 0.300 $\pm$ 0.028 | 0.365                | +0.045                           |           |
| D-token (prepended) | 0.269 $\pm$ 0.025 | 0.375                | +0.015                           |           |

Table 3. Exploratory main ablation (AITW all\_with\_coords,  $n_{\text{train}}=1200$ , 3 seeds, historical leaky splits). Episode-clustered  $\Delta$  vs. A with 95% cluster-bootstrap CIs and sign-flip permutation  $p$ . Under the repaired analysis, B survives (Holm-adjusted  $p = 0.017$  across the four vs.-A contrasts); D-hook is marginal after correction (Holm  $p = 0.068$ ); C and D-token are not significant.

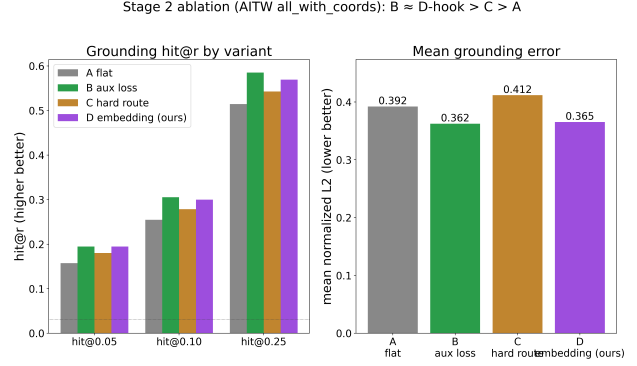


Figure 2. Exploratory ablation on all\_with\_coords (historical splits; per-seed values). Shown for continuity with the historical record; Table 3 carries the calibrated statistics and Table 5 the confirmatory test-set versions.

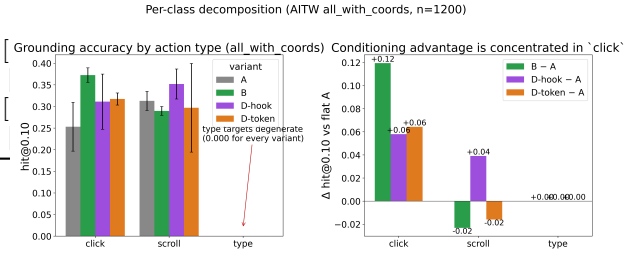


Figure 3. Exploratory hit@0.10 by action class (all\_with\_coords). The conditioning advantage concentrates in click; type is degenerate on AITW (all variants score exactly 0.000, mean L2 at the  $\approx \sqrt{2}$  corner-to-corner distance).

Stage-2 conditioning could matter at all.

## 6.2. Main ablation under clustered statistics

Table 3 re-scores the historical headline with the repaired statistics (per-seed values in Figure 2). Three observations. First, the auxiliary-loss effect survives: B-A on hit@0.10 is +0.051 [+0.020, +0.081] ( $p = 0.0043$ ), where the invalid pooled analysis had claimed  $p = 0.0002$  – the conclusion stands but the evidence was overstated by an order of magnitude. Second, D-hook weakens to marginal after multiplicity correction. Third, the comparison our proposal cared about most – D-token vs. B – is no longer significant ( $-0.036 [-0.077, +0.005]$ ,  $p = 0.114$ ): on the historical data we cannot even claim the prepended embedding is significantly worse than the auxiliary loss, only that it shows no sign of being better.

## 6.3. Per-class decomposition: the gain is click disambiguation

Breaking hit@0.10 down by class (Figure 3): B gains +0.120 on click (68% of examples) with a small scroll cost; C has the largest click gain (+0.122) but destroys scroll ( $-0.144$ ) by forcing a decode prefix that derails the gesture class; D-hook alone improves both (click +0.058, scroll +0.039). The type class scores exactly 0.000 across all variants with mean normalized L2 at or slightly above  $\sqrt{2}$ , the corner-to-corner distance (a few predictions land outside the unit square): AITW type events carry degenerate touch coordinates and are ungroundable by any model. This corrects our earlier “type $\rightarrow$ text-field localization” interpretation: the real signal is click-vs-scroll/type disambiguation for the precise-target class, and the “type” rows merely add a constant drag that cancels in deltas.

Data-scaling: action-type conditioning advantage vs training size  
(AITW all\_with\_coords; 300-1200 are 3-seed means, 2500/5000 single seed)

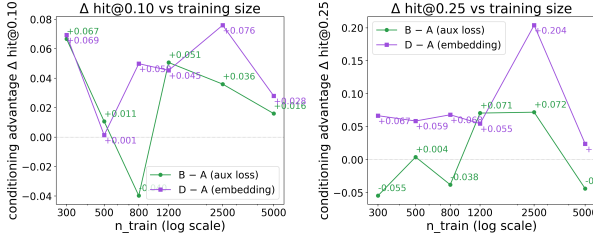


Figure 4. Exploratory conditioning advantage ( $\Delta$  hit@0.10 vs. A) across  $n_{\text{train}}$ . Caveat from Section 5.2: each historical size used a different val slice with different goal leakage (up to 85%), so this curve is directional at best; the confirmatory grid (Table 7) fixes one frozen test set for all sizes.

#### 6.4. Control conditions

On taps\_and\_swipes (action type spatially uninformative), the historical 3-seed means show no conditioned variant meaningfully above A (A  $0.390 \pm 0.008$ , D-hook  $0.392 \pm 0.035$  – a difference well inside one seed-std – with B  $0.368 \pm 0.009$  and C  $0.325 \pm 0.032$  at or below A on hit@0.10); per-example logs were not retained for A on this mix, so the clustered version of this control appears only in the confirmatory rerun (Table 6). On Multimodal-Mind2Web (click/type-dominated), D-hook vs. A was  $-0.006$  hit@0.10 /  $+0.018$  hit@0.25 over 2 seeds – a null, retained here deliberately: conditioning does not help where action type carries no spatial information, on either dataset.

#### 6.5. Low-data behavior: no clean curve

Our proposal predicted the advantage is a low-data prior. The historical grid ( $n \in \{300, 500, 800\}$ , 3 seeds) does not show a clean monotone: B-A on hit@0.10 is  $+0.067$  at  $n=300$  but  $+0.011$  at  $n=500$  and  $-0.040$  at  $n=800$ ; D-hook is more stable ( $+0.069$ ,  $+0.001$ ,  $+0.048$ ) and leads hit@0.25 and mean L2 at every low-data size. The two single-seed high-end cells disagree with each other: at  $n=5000$  the deltas shrink and partially reverse (B-A hit@0.25 goes negative), while at  $n=2500$  D-hook posts the largest conditioning deltas anywhere in the study ( $+0.076$  hit@0.10,  $+0.204$  hit@0.25) – single-seed noise on a differently-leaky val slice, cutting in the direction favorable to conditioning. We report all of this as: inconclusive on the low-data-prior hypothesis – noisy, non-monotone, and confounded by per-size val sets (Figure 4); the prespecified low-data claim is evaluated on the frozen test set in Section 7. An earlier draft of this work called the low-data prediction “confirmed”; that claim did not survive this audit.

| contrast     | $\Delta$ hit@0.10 [95% CI]  | $\Delta$ hit@0.25 [95% CI]  | perm.    |
|--------------|-----------------------------|-----------------------------|----------|
| wrong – gold | $-0.192$ $[-0.233, -0.149]$ | $-0.329$ $[-0.372, -0.284]$ | $\leq 9$ |
| zero – gold  | $-0.093$ $[-0.128, -0.059]$ | $-0.112$ $[-0.145, -0.079]$ | $\leq 9$ |

Table 4. Exploratory D-token sensitivity (episode-clustered, 3 seeds): evaluating the same trained model with wrong or zeroed action embeddings severely degrades grounding. This establishes the embedding is used at inference – a sensitivity/oracle analysis, not evidence that conditioning caused any gain over A.



Figure 5. Image-region attention for a held-out click example under three action-type conditionings (qualitative illustration; screenshot attention mass 0.204 under click vs. 0.132/0.110 under type/scroll).

#### 6.6. The e2e pipeline claim also weakens

The historical end-to-end result (predicted-type pipeline 0.271 vs. A 0.255 hit@0.10) was previously reported as a win. Under clustered statistics the paired difference is  $+0.016$   $[-0.017, +0.052]$  ( $p = 0.424$ ) – not significant; the oracle-type version is marginal ( $+0.037$   $[+0.005, +0.076]$ ). The deployable-pipeline question is therefore decided by the confirmatory run (P4, below), not the historical data.

#### 6.7. D-token causal-use sensitivity

Feeding the trained D-token model a wrong action id collapses grounding, and zeroing the embedding hurts substantially (Table 4). The refutation of the proposal’s mechanism is therefore precise: the prepended embedding is causally used, it is simply not the best way to exploit action-type supervision. Figure 5 shows one example where correct conditioning concentrates more attention mass on the screenshot; across the three examples we probed the pattern is not uniform, so we present it strictly as illustration, not evidence.

### 7. Confirmatory Results (untouched test, prespecified)

All variants were retrained from scratch on the frozen goal-disjoint splits (identical hyperparameters, seeds 42–44) and evaluated once on the untouched 600-step test set; the four primary contrasts, the headline

PENDING: confirmatory rerun results not yet rendered

Table 5. Confirmatory main ablation on the untouched, goal-disjoint AITW test set ( $n=600$  steps;  $n_{\text{train}}=1200$ ; 3 seeds; mean $\pm$ std across seeds).  $\Delta$  vs. A: episode-clustered 95% CI and permutation  $p$  (Holm-adjusted across the four primary contrasts where applicable).

PENDING: confirmatory rerun results not yet rendered

Table 6. Confirmatory control mix (taps\_and\_swipes) on its frozen 400-step test set: when action type is not spatially predictive, conditioning should not help; this is the falsifiable half of the thesis.

PENDING: confirmatory rerun results not yet rendered

Table 7. Confirmatory low-data grid on the same frozen test set for every training size (nested training prefixes), removing the per-size-val-set confound of Figure 4.

metric (hit@0.10), the clustered test, and Holm correction were committed before any test evaluation ran (Section 3.2).

P1 (auxiliary supervision). B vs. A on hit@0.10: [PENDING], permutation  $p$  = [PENDING], Holm  $p$  = [PENDING]; mean-L2 delta [PENDING] ( $p$  = [PENDING]).

P2 (inference-time conditioning). D-hook vs. A: [PENDING] (Holm  $p$  = [PENDING]).

P3 (the proposal’s mechanism). D-token vs. B: [PENDING] (Holm  $p$  = [PENDING]).

P4 (deployable pipeline). End-to-end predicted-type pipeline vs. A: [PENDING] (Holm  $p$  = [PENDING]); Stage-1 test accuracy [PENDING], oracle ceiling [PENDING] vs. predicted [PENDING] (paired gap [PENDING]).

Secondary contrasts, per-class decompositions, parse rates, and per-seed values for every run are in the released results/phase9\_rerun/rerun\_analysis.json; per-seed means for each primary contrast appear alongside the released analysis, and no procedure-level significance claim is made from three seeds (Section 5.3).

## 8. Limitations

Offline step-level hit@radius is not long-horizon task success: it scores logged single steps, without interaction, recovery, or compounding-error dynamics – the motivating failure mode is measured here only via its step-level signature. The backbone is a single 2B model; conclusions are about this scale and training recipe, not about GUI agents generally, and our base-

lines are small matched-compute models trained by us, not state-of-the-art systems. The confirmatory estimand is fixed checkpoints, not the training procedure (three seeds; per-seed effects reported). AITW’s type coordinates are degenerate, so “grounding” on all\_with\_coords is effectively click/scroll grounding plus a constant; the click-concentration analysis is post-hoc on exploratory data and only its prespecified aggregate versions are confirmatory. Goal strings are our dependence proxy; 17 near-duplicate train/test goal pairs (disclosed in the manifest) survive normalization, and subtler sharing (same app, similar screens) is not modeled. The Mind2Web control used 2 seeds without per-example logs. Finally, the mirror we pin is a third-party repackaging of AITW; we verified revision stability but not byte-level provenance against the original release.

## 9. Conclusion

The broad factorization thesis survives careful evaluation in narrowed form: auxiliary action-type supervision helps a small GUI grounding model where action type is spatially predictive, the effect concentrates in click disambiguation, and it disappears on matched controls – with the confirmatory test-set verdicts for all four prespecified contrasts reported above exactly as they came out. The specific architecture our proposal advocated – a learned prepended action embedding – is refuted as the winning mechanism despite being demonstrably used by the model. En route we hit, diagnosed, and repaired three evaluation artifacts (a silent M-RoPE position bug producing an  $8\sigma$  false negative; goal-level split leakage reaching 85%; pseudo-replicated statistics with 21.2% measured type-I error), each individually capable of flipping the paper’s conclusion. We suspect none of them is unique to us: stream-sliced splits, embedding-injection shortcuts, and pooled per-seed bootstraps are all common patterns. The frozen splits, per-example predictions, clustered-inference code, and prespecification are released so both the object-level result and the audit are reproducible.

## Contributions

A. Chauhan: training and evaluation infrastructure, Stage-2 conditioning variants, cloud experiment orchestration, statistical reanalysis and confirmatory protocol, writing. A. Ilyasov: Stage-1 classifier study, data pipeline and error analysis, figures, editing. Both authors contributed to problem formulation and experimental design and approved the final manuscript.

## Generative AI Usage Statement

Large-language-model assistants (Anthropic Claude) were used extensively under the authors' direction: writing and debugging training, evaluation, and statistical-analysis code; orchestrating cloud experiments; drafting the integrity audit that led to Sections 5 and 7; and drafting and editing this manuscript. The authors specified the research questions and methodology, reviewed the analyses, verified reported numbers against the committed result artifacts (every table is generated programmatically from those artifacts), and take full responsibility for all content.

## References

- [1] M. Ahn et al. Do as I can, not as I say: Grounding language in robotic affordances. In CoRL, 2022.
- [2] Anonymous. RegionFocus: Training-free visual test-time scaling for GUI grounding. arXiv preprint, 2025.
- [3] A. Brohan et al. RT-2: Vision-language-action models transfer web knowledge to robotic control. arXiv preprint arXiv:2307.15818, 2023.
- [4] K. Cheng et al. SeeClick: Harnessing GUI grounding for advanced visual GUI agents. arXiv preprint arXiv:2401.10935, 2024.
- [5] X. Deng et al. Mind2Web: Towards a generalist agent for the web. In NeurIPS, 2023.
- [6] B. Gou et al. Navigating the digital world as humans do: Universal visual grounding for GUI agents. arXiv preprint arXiv:2406.01729, 2024.
- [7] K. Q. Lin et al. ShowUI: One vision-language-action model for GUI visual agent. arXiv preprint arXiv:2411.17465, 2024.
- [8] Z. Lu et al. UI-R1: Enhancing action prediction of GUI agents by reinforcement learning. arXiv preprint arXiv:2503.21620, 2025.
- [9] C. Rawles et al. Android in the wild: A large-scale dataset for android device control. In NeurIPS, 2023.
- [10] P. Wang et al. Qwen2-VL: Enhancing vision-language model's perception of the world at any resolution. arXiv preprint arXiv:2409.12191, 2024.
- [11] Z. Wu et al. OS-Atlas: A foundation action model for generalist GUI agents. arXiv preprint arXiv:2410.23218, 2024.
- [12] S. Yao et al. ReAct: Synergizing reasoning and acting in language models. arXiv preprint arXiv:2210.03629, 2022.
- [13] Y. Yao et al. UI-TARS: Pioneering automated GUI interaction with native agents. arXiv preprint arXiv:2501.12326, 2025.
- [14] Q. Zheng et al. GUI-Actor: Coordinate-free visual grounding for GUI agents. arXiv preprint arXiv:2503.15197, 2025.